

Geometry of Particle Methods

We can also allow *weighted* particles:

$$\hat{\mu}_t := \sum_{i=1}^n w(t, X_i) \delta_{X^{(i)}} \quad \{X^{(i)}\}_{i=1\dots n}$$

Dynamics:

$$\partial_t w(t, X) = \alpha(t, X, \mu_t) \quad \Longleftrightarrow \quad \partial_t \mu(t, X) = \alpha(t, X, \mu_t) \mu(t, X, \mu_t)$$

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Definition (Fisher-Rao Gradient)

The Fisher-Rao gradient of $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$ at the point $\mu \in \mathcal{P}_2(\mathcal{M})$ is (when it exists and is square integrable) the function,

$$\nabla^{FR} \mathcal{F}[\mu](x) = \delta \mathcal{F}[\mu](x) - \int \delta \mathcal{F}[\mu](x) \mu(dx)$$

Intuition: steepest descent dynamics via mass reweighting.